

江苏省 2015 年普通高校专转本统一考试

高等数学 答案解析

一、选择题

1. 解: 因为 $\lim_{x \rightarrow 0} \frac{1 - e^{\sin x}}{x} = -\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x} = -\lim_{x \rightarrow 0} \frac{\sin x}{x} = -1$

所以当 $x \rightarrow 0$ 时, $f(x)$ 与 $g(x)$ 是同阶无穷小

2. 解: $y = (1-x)^x = e^{x \ln(1-x)}$

$$y' = e^{x \ln(1-x)} \left[\ln(1-x) + x \cdot \frac{-1}{1-x} \right] = (1-x)^x \left[\ln(1-x) - \frac{x}{1-x} \right]$$

$$dy = y' dx = (1-x)^x \left[\ln(1-x) - \frac{x}{1-x} \right] dx$$

3. 解: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}} + 1}{e^{\frac{1}{x}} - 1} = -1$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x}} + 1}{e^{\frac{1}{x}} - 1} = 1$$

所以 $x=0$ 是 $f(x)$ 的跳跃间断点

4. 解: 因为 $F(x)$ 是 $f(x)$ 的一个原函数, 所以 $\int f(x) dx = F(x) + c$

$$\text{于是 } \int f(3-2x) dx = -\frac{1}{2} \int f(3-2x) d(3-2x) = -\frac{1}{2} F(3-2x) + c$$

5. 解: 对于 A, $\sum_{n=1}^{\infty} \frac{(-1)^n - n}{n^2} = \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{n^2} - \frac{1}{n} \right]$

因为 $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ 收敛, $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 所以级数发散

对于 B, 因为 $\lim_{n \rightarrow \infty} \frac{n+1}{2n-1} = \frac{1}{2} \neq 0$, 所以级数发散

$$\text{对于 C, } \sum_{n=1}^{\infty} \left| (-1)^n \frac{n!}{n^n} \right| = \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

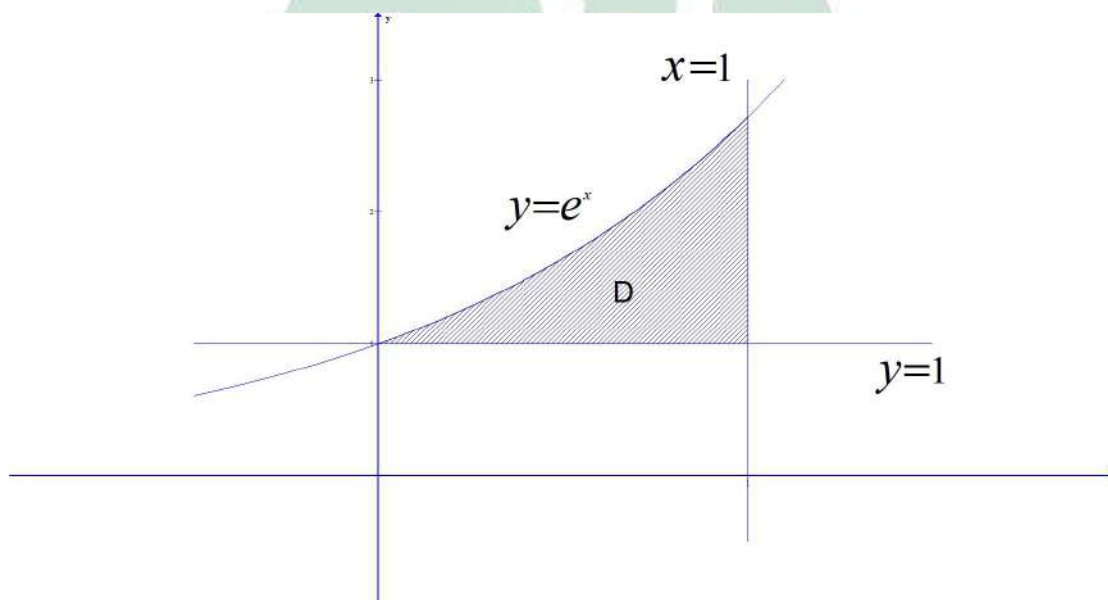
$$\text{因为 } \lim_{n \rightarrow \infty} \frac{u_{n+1}(x)}{u_n(x)} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1$$

由比值判别法可知 $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ 收敛, 故 $\sum_{n=1}^{\infty} (-1)^n \frac{n!}{n^n}$ 绝对收敛

对于 D, 因为 $\sum_{n=1}^{\infty} \left| (-1)^n \frac{n+1}{n^2} \right| = \sum_{n=1}^{\infty} \frac{n+1}{n^2}$ 发散, 而 $\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{n^2}$ 收敛, 所以条件收敛

6. 解: 积分区域 $D: \begin{cases} \ln y \leq x \leq 1 \\ 1 \leq y \leq e \end{cases}$

交换积分次序得 $\int_1^e dy \int_{\ln y}^1 f(x, y) dx = \int_0^1 dx \int_1^{e^x} f(x, y) dy$



二、填空题

7. 解: 因为 $\lim_{n \rightarrow \infty} \left(1 - \frac{x}{n} - 1 \right) \cdot n = -x$

所以 $f(x) = \lim_{n \rightarrow \infty} \left(1 - \frac{x}{n} \right)^n = e^{-x}$, 于是 $f(\ln 2) = e^{-\ln 2} = \frac{1}{2}$

8. 解: $\frac{dx}{dt} = 3t^2 - 2$, $\frac{dy}{dt} = 3t^2$

$$\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{3t^2}{3t^2 - 2}$$

点(0,2)对应得参数为 $t=1$, 所以 $\left.\frac{dy}{dx}\right|_{t=1} = \frac{3t^2}{3t^2-2}\bigg|_{t=1} = 3$

切线方程为 $y-2=3x$, 即 $y=3x+2$

9. 略

10. 解: $f'(x) = \frac{-1}{(2x+1)^2} \cdot 2$, $f''(x) = \frac{1 \cdot 2}{(2x+1)^3} \cdot 2^2$

$f'''(x) = \frac{-1 \cdot 2 \cdot 3}{(2x+1)^4} \cdot 2^3$, $f^{(4)}(x) = \frac{1 \cdot 2 \cdot 3 \cdot 4}{(2x+1)^5} \cdot 2^4$

所以 $f^{(n)}(x) = \frac{(-1)^n \cdot n! \cdot 2^n}{(2x+1)^{n+1}}$

11. 解: 方程可化成 $y' + \frac{-1}{x}y = x$

通解 $y = e^{\int \frac{1}{x} dx} \left[\int x e^{-\int \frac{1}{x} dx} dx + c \right] = e^{\ln x} \left(\int x e^{-\ln x} dx + c \right)$

$= x \left(\int dx + c \right) = x(x+c) = x^2 + cx$

将 $y|_{x=1} = 2$ 代入通解得 $c=1$, 所以 $y = x^2 + x$

12. 解: 令 $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x-1)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{2^n(x-1)^n} \right| = 2|x-1| < 1$

解得 $\frac{1}{2} < x < \frac{3}{2}$

当 $x = \frac{1}{2}$ 时, 级数为 $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ 收敛, 当 $x = \frac{3}{2}$ 时, 级数为 $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ 发散

所以收敛域为 $\left[\frac{1}{2}, \frac{3}{2} \right)$

三、计算题

13. 解: $\lim_{x \rightarrow 0} \frac{\int_0^x t \arcsin t dt}{2e^x - x^2 - 2x - 2} = \lim_{x \rightarrow 0} \frac{x \arcsin x}{2e^x - 2x - 2} = \lim_{x \rightarrow 0} \frac{x^2}{2e^x - 2x - 2}$

$$= \lim_{x \rightarrow 0} \frac{2x}{2e^x - 2} = \lim_{x \rightarrow 0} \frac{x}{e^x - 1} = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

14. 解: 当 $x \neq 0$ 时, $f'(x) = \frac{(1 - \cos x) \cdot x^2 - (x - \sin x) \cdot 2x}{x^4} = \frac{2 \sin x - x - x \cos x}{x^3}$

当 $x = 0$ 时, $f'(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \frac{1}{6}$

$$\text{所以 } f'(x) = \begin{cases} \frac{2 \sin x - x - x \cos x}{x^3} & x \neq 0 \\ \frac{1}{6} & x = 0 \end{cases}$$

15. 略

16. 解: $\int \frac{x^3}{\sqrt{9-x^2}} dx = \frac{1}{2} \int \frac{9-x^2-9}{\sqrt{9-x^2}} d(9-x^2)$

$$= \frac{1}{2} \int \sqrt{9-x^2} d(9-x^2) - \frac{9}{2} \int \frac{1}{\sqrt{9-x^2}} d(9-x^2)$$

$$= \frac{1}{3} (9-x^2) \sqrt{9-x^2} - 9 \sqrt{9-x^2} + c$$

17. 解: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^2 + x) \sin x dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \sin x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin x dx$

$$= 0 + 2 \int_0^{\frac{\pi}{2}} x \sin x dx = -2 \int_0^{\frac{\pi}{2}} x d \cos x$$

$$= -2x \cos x \Big|_0^{\frac{\pi}{2}} + 2 \int_0^{\frac{\pi}{2}} \cos x dx = 2 \sin x \Big|_0^{\frac{\pi}{2}} = 2$$

18. 解: $\frac{\partial z}{\partial x} = \frac{1}{y} f_1' + \varphi'(x) f_2'$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-1}{y^2} f_1' + \frac{1}{y} \left[f_{11}'' \cdot \frac{-x}{y^2} + f_{12}'' \cdot 0 \right] + \varphi'(x) \left[f_{21}'' \cdot \frac{-x}{y^2} + f_{22}'' \cdot 0 \right]$$

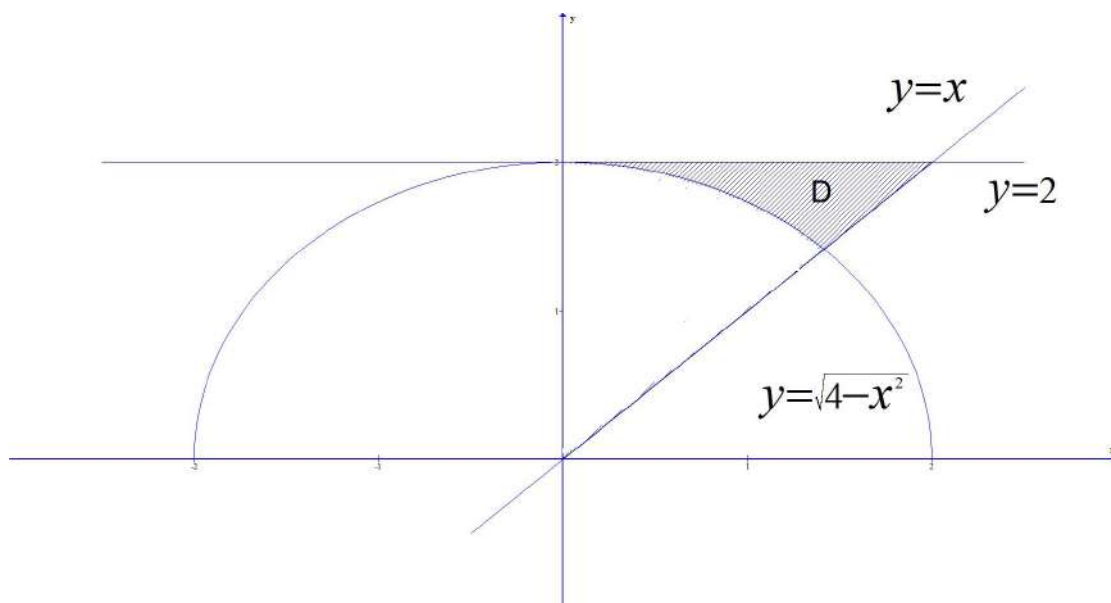
$$= \frac{-1}{y^2} f_1' - \frac{x}{y^3} f_{11}'' - \frac{x}{y^2} \varphi'(x) f_{21}''$$

19. 解: $\iint_D xy dx dy = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_2^{\frac{2}{\sin \theta}} r \cos \theta \cdot r \sin \theta \cdot r dr = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \cdot \cos \theta \cdot \sin \theta \cdot \frac{1}{4} r^4 \Big|_2^{\frac{2}{\sin \theta}}$

$$= 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos \theta}{\sin^3 \theta} d\theta - 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos \theta \sin \theta d\theta$$

$$= 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sin^3 \theta} d \sin \theta - 4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin \theta d \sin \theta$$

$$= -2 \frac{1}{\sin^2 \theta} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} - 2 \sin^2 \theta \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 1$$



20. 解：由微分方程解的结构定理可知 $\bar{y} = c_1 e^x + c_2 e^{2x}$ 是微分方程 $y'' + py' + qy = 0$ 的通解

故此方程的特征方程为 $(r-1)(r-2)=0$ ，即 $r^2 - 3r + 2 = 0$

对应的齐次方程为 $y'' - 3y' + 2 = 0$ ，所以 $p = -3, q = 2$

又因为 $y^* = xe^{3x}$ 是微分方程 $y'' - 3y' + 2 = f(x)$ 的特解

所以应有 $(xe^{3x})'' - 3(xe^{3x})' + 2xe^{3x} = f(x)$ ，所以 $f(x) = (2x+3)e^{3x}$

所求微分方程为 $y'' - 3y' + 2 = (2x+3)e^{3x}$

四、综合题

21. 解：(1) $V_x = \pi \int_0^a (ax)^2 dx - \pi \int_0^a (x^2)^2 dx = \pi a^2 \cdot \frac{1}{3} x^3 \Big|_0^a - \frac{1}{5} \pi x^5 \Big|_0^a$

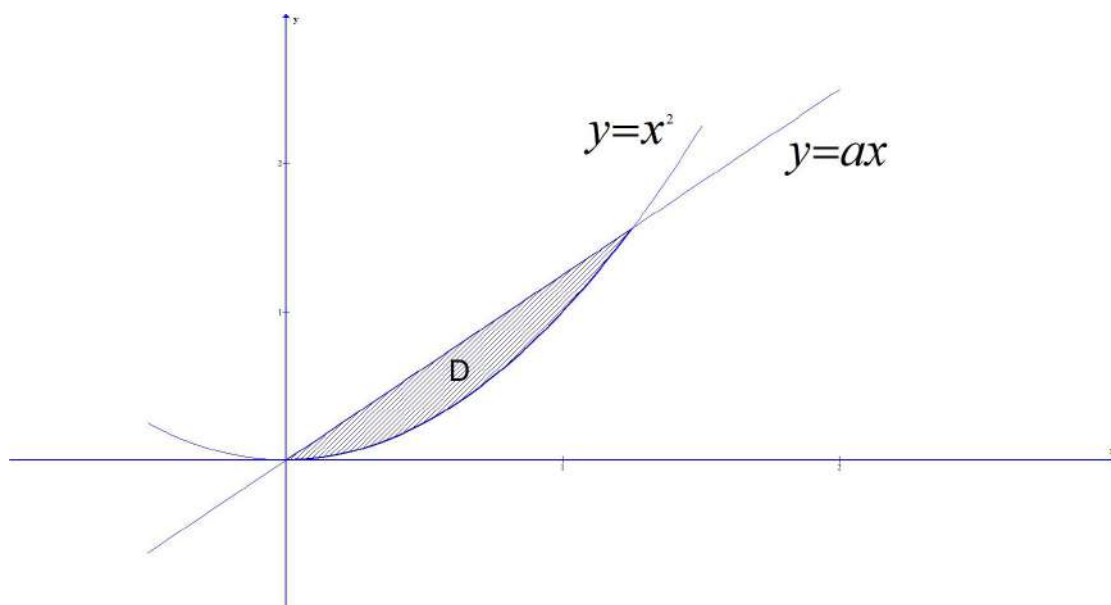
$$= \frac{\pi a^5}{3} - \frac{\pi a^5}{5} = \frac{2\pi a^5}{15}$$

$$V_y = \pi \int_0^{a^2} (\sqrt{y})^2 dy - \pi \int_0^{a^2} \left(\frac{y}{a}\right)^2 dy = \frac{1}{2} \pi y^2 \Big|_0^{a^2} - \frac{\pi}{a^2} \cdot \frac{1}{3} y^3 \Big|_0^{a^2}$$

$$= \frac{1}{2} \pi a^4 - \frac{1}{3} \pi a^4 = \frac{1}{6} \pi a^4$$

根据题意, 有 $V_x = V_y$, 所以 $\frac{2}{15} \pi a^5 = \frac{1}{6} \pi a^4$, 解得 $a = \frac{5}{4}, a = 0$ (舍去)

$$(2) S = \iint_D dx dy = \int_0^{\frac{5}{4}} dx \int_{x^2}^{\frac{5}{4}x} dy = \int_0^{\frac{5}{4}} \left(\frac{5}{4}x - x^2 \right) dx = \left(\frac{5}{8}x^2 - \frac{1}{3}x^3 \right) \Big|_0^{\frac{5}{4}} = \frac{125}{384}$$



22. 解: (1) $f'(x) = \frac{a(x+1)^2 - 2(ax+b)(x+1)}{(x+1)^4} = \frac{-ax + a - 2b}{(x+1)^3}$

因为 $f(x)$ 在 $x=1$ 处取得极值 $-\frac{1}{4}$

所以应有 $\begin{cases} f'(1) = 0 \\ f(1) = -\frac{1}{4} \end{cases}$, 即 $\begin{cases} \frac{-2b}{8} = 0 \\ \frac{a+b}{4} = -\frac{1}{4} \end{cases}$, 解得 $a = -1, b = 0$

(2) $f'(x) = \frac{x-1}{(x+1)^3}$, $f''(x) = \frac{-2x+4}{(x+1)^4}$, 令 $f''(x) = 0$ 得 $x = 2$

x	$(-\infty, -1)$	-1	$(-1, 2)$	2	$(2, +\infty)$
$f''(x)$	$+$	无定义	$+$	0	$-$
$f(x)$	$\cup$	无定义	$\cup$	拐点	$\cap$

由表可知, 曲线的凹区间为 $(-\infty, -1)$, $(-1, 2)$, 凸区间为 $(2, +\infty)$

$$\text{拐点} \left(2, -\frac{2}{9} \right)$$

$$(3) f(x) = \frac{-x}{(x+1)^2}$$

因为 $\lim_{x \rightarrow +\infty} \frac{-x}{(x+1)^2} = 0$, 所以 $y = 0$ 是曲线的水平渐近线

因为 $\lim_{x \rightarrow -1} \frac{-1}{(x+1)^2} = \infty$, 所以 $x = -1$ 是曲线的垂直渐近线

五、证明题

22. 证明: 令 $F(x) = (x-2)\ln(1-x) - 2x$, 则 $F(0) = 0$

$$F'(x) = \ln(1-x) + \frac{-(x-2)}{1-x} - 2 = \ln(1-x) + \frac{x-2}{x-1} - 2, \text{ 则 } F'(0) = 0$$

$$F''(x) = \frac{-1}{1-x} + \frac{1}{(x-1)^2} = \frac{x}{(x-1)^2}$$

当 $0 < x < 1$ 时, $F''(x) > 0$, 所以 $F'(x)$ 单调递增, $F'(x) > F'(0) = 0$

由 $F'(x) > 0$ 又可知 $F(x)$ 单调递增, $F(x) > F(0) = 0$

即 $(x-2)\ln(1-x) - 2x > 0$, 所以 $(x-2)\ln(1-x) > 2x$

24. 证明: 方程两边对 x 求偏导数得 $\frac{\partial z}{\partial x} = f(y^2 - z^2) + xf'(-2z)\frac{\partial z}{\partial x}$

$$\text{所以 } \frac{\partial z}{\partial x} = \frac{f(y^2 - z^2)}{1 + 2xzf'}$$

$$\text{方程两边对 } y \text{ 求偏导数得 } 1 + \frac{\partial z}{\partial y} = xf' \left(2y - 2z \frac{\partial z}{\partial x} \right)$$

$$\text{所以 } \frac{\partial z}{\partial y} = \frac{2xyf' - 1}{1 + 2xzf'}$$

$$\begin{aligned}
 \text{于是 } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= \frac{xf(y^2 - z^2)}{1 + 2xyzf'} + \frac{2xyzf' - z}{1 + 2xyzf'} = \frac{xf(y^2 - z^2) + 2xyzf' - z}{1 + 2xyzf'} \\
 &= \frac{y + 2xyzf'}{1 + 2xyzf'} = \frac{y(1 + 2xyzf')}{1 + 2xyzf'} = y
 \end{aligned}$$



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